Nhan Nguyen

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Education

California State University, Long Beach — **Long Beach, CA**
B.S. Computer Science, Minor in Statistics — *Expected Graduation: May 2027*

- **Relevant Coursework:** Data Structures, Algorithms, Object-Oriented Programming, Software Engineering, Operating Systems, Databases, Machine Learning, Artificial Intelligence, Deep Learning

Projects

OrderFlow — **https://github.com/Detims/OrderFlow**

- Developed separate order and inventory **microservices** in **Java** with **Spring Boot**, exposing **REST APIs** backed by independent PostgreSQL databases.
- Designed an asynchronous order workflow with **AWS SQS** request and result queues, ensuring up-to-date order statuses based on inventory results with idempotent message handling.
- Built a local development environment with **Docker** and **LocalStack**, and used **JUnit** and **Mockito** unit tests to validate reservation outcomes and message redelivery.
- Configured **Jenkins** CI to build the **Maven** projects and run automated tests; containerized the services and deployed them to a local **Kubernetes** cluster.

CloudVault — **https://github.com/Detims/CloudVault**

- Developed a secure document management application using **Java**, **Spring Boot**, and **Google OpenID Connect** authentication, enforcing user-specific access controls through **Spring Security**.
- Integrated private **AWS S3** object storage with presigned upload/download URLs and **Cassandra**-backed document data, implementing stateful upload confirmation and ownership validation.
- Containerized application services and deployed them to local **Kubernetes**, automating builds, testing, image publishing, and approved deployments using **GitHub Actions**.

Fire Detection System

- Developed an **IoT**-based fire detection system integrating flame, CO2, and smoke sensors, streaming real-time data to **NeonDB** for continuous monitoring and analysis of environmental risk factors.
- Engineered a data pipeline with **Pandas** and **Scikit-learn** to preprocess and clean a high-frequency sensor dataset of **50,000+** records, utilizing moving averages with rolling windows to enable accurate feature engineering and training.
- Trained and evaluated **Random Forest**, **Logistic Regression** and **Support Vector Machine** classifiers, achieving **96%** accuracy, **90%** precision, and **0.67** F1 score with the selected Random Forest model.

InboxAssist — **https://github.com/magichouse21/InboxAssist**

- Collaborated on an **Azure**-based **full stack** Outlook browser extension, integrating **LLM**-powered email summarization, semantic search, question answering, and assisted drafting.
- Implemented Microsoft **OAuth** authentication and session state management, supporting authenticated access to email data.
- Optimized email parsing techniques for unread email summarization by restructuring **Microsoft Graph API** data processing around message body previews, reducing LLM token utilization by **87%**.
- Refactored retrieval functionality by replacing **FAISS**-based processing with Javascript chunking and vector embeddings.

Experience

Machine Learning Engineer — **June 2026 – August 2026**
American Society of Mechanical Engineers — *Long Beach, CA*

- Developed components of a Raspberry Pi 4-based computer vision pipeline for an autonomous glider, integrating **OpenCV** for object detection, sensor measurements, and GPS/gyroscope telemetry.
- Trained and evaluated a **YOLOv8n** object detection model using **ClearML** for experiment tracking and model management, achieving F1 scores of **0.77** and **0.87** for person and building detection across a custom **15,000+** image dataset..
- Enhanced autonomous flight analysis by building the ground control interface with **Flask** and picamera to conduct object detection and stream video capture information, with an average latency of approximately **300** ms.

Web Developer — **August 2025 – June 2026**
Union of Vietnamese Student Associations of Southern California — *Garden Grove, CA*

- Developed and maintained features for a large-scale community platform using **React**, **TypeScript**, and **Tailwind CSS**, implementing responsive layouts and interactive UI components alongside various site-wide features.
- Debugged, resolved, and optimized various frontend issues involving component behavior and mobile responsiveness, standardizing component usage and quickened load times by up to **3x**.
- Collaborated in an iterative **Agile** development environment with weekly sprints, delivering new features and bug fixes through **Git**-based version control and peer-reviewed pull requests under 1-2 week deadlines.

Technical Skills

Languages: Java, Python, Javascript, Typescript, C++, HTML/CSS, SQL
Frameworks/Libraries: Spring Boot, REST APIs, Flask, Express.js, Node.js, React, Tailwind CSS, Pandas, Scikit-learn, PyTorch
Technologies: Docker, Kubernetes, AWS, Jenkins, LocalStack, JUnit, Mockito, Azure, PostgreSQL, GitHub Actions, Cassandra